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Individual Differences Link Sensory Processing and Motor Control

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Research on saccadic and pursuit eye movements led to great advances in our understanding of sensorimotor processing and human behavior. However, studies often have focused on isolated saccadic and pursuit eye movements measured with respect to different sensory information (static vs. dynamic targets). Here, we leveraged interindividual differences across a carefully balanced combination of different tasks to demonstrate that critical links in the control of oculomotor behavior were previously missed. We observed correlations in eye movement behavior across tasks, but only when compared with the same sensory information (e.g., pursuit gain and accuracy of saccades to moving targets). Within the same task, the coordination of saccadic and pursuit eye movements was tailored to the strengths of the individual: observers with more accurate saccades to moving targets rely on them more to catch up with moving targets. Our results have profound implications for the theoretical understanding of sensorimotor processing for oculomotor control. They necessitate a reevaluation of previous data used to map brain circuits for saccadic and pursuit eye movements measured with different types of relevant sensory information. Additionally, they underscore the importance of moving beyond average observations to embrace individual differences as a rich source of information. These individual differences not only reveal the strengths and weaknesses of observers. When combined across different tasks, they allow insights about *why* observers behave differently in a given task.

Keywords: individual differences, eye movements, saccades, pursuit, sensory processing

Due to the foveal-centric organization of our visual system, eye movements are an essential part of perception and are studied across a wide range of different research areas (for overviews, see Klein & Ettinger, 2019; Kowler et al., 2019; Krauzlis, 2005; Lisberger, 2015; Schütz et al., 2011). Humans mainly use a combination of fast saccadic and slow pursuit eye movements to keep targets of interest within foveal vision. Because these two types of eye movements operate in distinct speed ranges, they have been considered fully independent. This has also led to them being studied separately with paradigms that maximize the differences. Most lines of saccadic research are conducted with static visual input. For example, isolated saccades are studied when looking at simple stationary dots (Bahill et al., 1975; Munoz & Everling, 2004; Pélisson et al., 2010), during reading of static text (e.g., Engbert et al., 2005; Rayner, 1998) or gaze distributions are used to predict where observers fixate in static images (e.g., Itti et al., 1998; Kümmerer et al., 2016, 2022). In contrast, pursuit eye movement research makes almost exclusive use of dynamic, moving stimuli. For example, pursuit is mostly studied with moving dots to understand how the brain transforms visual input into a continuous motor output (Lisberger, 2015) or to study the link between motion perception and pursuit behavior (Spering & Montagnini, 2011). Such pursuit studies often rely on a specific paradigm, the step ramp paradigm (Rashbass, 1961), to reduce the need for naturally occurring catch-up saccades.

The focus on different paradigms also leads to different sensory control signals: for static targets, the main signal driving the eye movement is position-related (the mismatch between eye and target position), and for dynamic targets, the main signal is velocity-related

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All data are available in the main text. Raw data for each of the individual tasks and preregistration of the study design and data analysis are available at PsychArchives at https://doi.org/10.23668/psycharchives.12502. The analysis code is publicly available on GitHub (https://github.com/AlexanderGoettker/IndDiffSensoryMotor/).

Alexander Goettker played a lead role in conceptualization, formal analysis, methodology, project administration, software, validation, visualization, writing–original draft, and writing–review and editing and a supporting role in funding acquisition. Karl R. Gegenfurtner played a lead role in funding acquisition, a supporting role in conceptualization, and an equal role in writing–review and editing.

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(the mismatch between eye and target velocities). This leads to an important but often neglected confound when trying to understand oculomotor control: The relevant sensory information (position vs. velocity) is coupled with the measured oculomotor behavior (saccade vs. pursuit). Due to this confound inherent in commonly used paradigms, it is impossible to dissociate whether differences in the control of saccadic and pursuit eye movements are based on the oculomotor response or whether they are based on differences in the related sensory signals. This is also relevant for comparing brain responses for saccades to static targets with pursuit of moving targets, which has been a common tool for mapping out oculomotor brain circuits (Agtzidis et al., 2020; Petit et al., 1997; Rosano et al., 2002). Here, we aim to solve this confound by combining two lines of research that break the typical study of the average behavior of isolated eye movements.

First, while often studied in isolation, during natural behavior, saccadic and pursuit eye movements often occur together to allow for optimal tracking. Interactions and shared information between saccades and pursuit have been shown for complex natural movements in highly dynamic situations (Dorr et al., 2010; Goettker et al., 2021; Land & McLeod, 2000), but also with simpler paradigms under full experimental control (de Brouwer et al., 2002; Goettker, Braun, et al., 2019; Liston & Krauzlis, 2003, 2005; Orban de Xivry et al., 2006; for reviews, see Goettker & Gegenfurtner, 2021; Orban de Xivry & Lefèvre, 2007). It has also been shown that the neurophysiological structures involved in the control of saccades and pursuit are not as independent as initially thought but exhibit a large overlap (Krauzlis, 2004, 2005). In line with this overlap, converging evidence demonstrates that both saccades and pursuit are used and able to correct for both position- and velocity-related signals (Orban de Xivry & Lefèvre, 2007): Saccades to moving targets successfully integrate velocity-information (Fleuriet et al., 2011; Goettker, Braun, et al., 2019) and pursuit is affected by target position (Blohm et al., 2005; Buonocore et al., 2019; Pola & Wyatt, 1980; Segraves & Goldberg, 1994).

Second, the vast majority of eye movement studies focus on the average behavior across many observers, and variance between observers is often treated as noise. However, massive individual differences have been shown for the control of both saccadic (Castelhano & Henderson, 2008; Constantinidis et al., 2003; Constantino et al., 2017; de Haas et al., 2019; Kennedy et al., 2017) and pursuit eye movements (Lenzenweger & O'Driscoll, 2006; Schröder et al., 2021; Versino et al., 1993), and in how saccadic and pursuit eye movements are combined with tracking a moving target: The same stimulus that is tracked with smooth pursuit by one observer, is tracked by a combination of saccadic and pursuit eye movements by another observer (Goettker, Brenner, et al., 2019; Goettker et al., 2018). These interindividual differences in oculomotor behavior are strong and stable so that even ideas of an "oculomotor signature" or "oculomotor fingerprint" have been proposed (Bargary et al., 2017; de Haas et al., 2019; Lohr et al., 2022; Rigas et al., 2016). The focus on average behavior, therefore, might miss out on a substantial amount of interesting information.

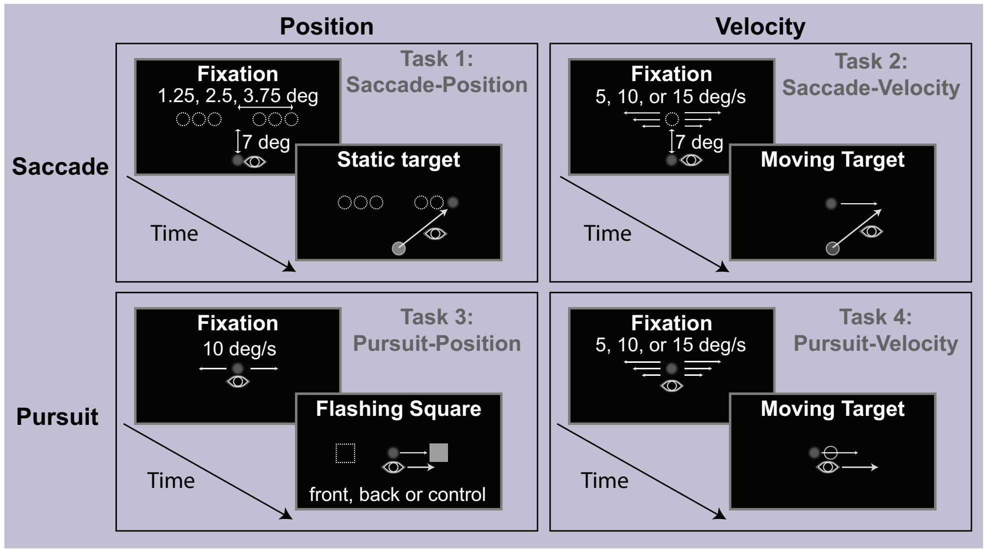
To tackle the often present confound of sensory information and measured oculomotor control, we will measure individual differences in saccades, pursuit, and saccade–pursuit interactions within the same group of observers. However, we not only measure individual variability but leverage individual differences to gain insight into the structure of sensory processing and oculomotor control (Wilmer & Nakayama, 2007, for reviews of more examples in vision science, see Mollon et al., 2017; Wilmer, 2008). We compared oculomotor performance across observers at two levels: First, we resolved the confound of stationary and moving stimuli in eye movement research with a unique combination of tasks (see Figure 1), where, for the same observers, we carefully balance sensory information (position, velocity) and the executed eye movement (saccade, pursuit). For example, we compared the accuracy of saccades to stationary targets with pursuit gain and contrasted this to the correlation between the accuracy of saccades to moving targets and pursuit gain. Second, within the same task with matched sensory information, we studied how saccadic and pursuit eye movements are coordinated by different individuals with different abilities in performing isolated saccades and pursuit. For example, will an observer who is making accurate saccadic eye movements use them more frequently when tracking a moving target?

Our results show that the behavior of observers for isolated saccadic and pursuit eye movements is correlated, but only across tasks where sensory information is matched (e.g., saccades and pursuit to moving targets). Thus, performance across tasks mainly seems to vary based on whether position or velocity information is relevant and less so on whether a saccade or pursuit is executed. Within the same task, we observed that the coordination of saccadic and pursuit eye movements is tailored to the individual strengths of observers. Observers with more accurate pursuit eye movements rely more on a pursuit to track a moving target, whereas observers with more accurate saccades to moving targets trigger saccades more frequently. We discuss how these results lead to a more integrative view of the oculomotor system and demonstrate the importance of considering individual differences in behavior. They are crucial for the reinterpretation of now classic results, comparing the control of saccadic eye movements and pursuit eye movements with either static or moving stimuli.

Results

A large group of observers ($N = 50$) completed a battery of different oculomotor tasks (see Figure 1). To address the typical confound of saccadic eye movements being studied with static stimuli and pursuit eye movements with moving stimuli, we balanced the relevant sensory information (position- or velocity-related signals) across eye movements in our tasks (see the Method section for a detailed description of the tasks). Saccade performance was measured to static and moving targets. Participants initially fixated at the bottom of the screen, and then a target was shown on the vertical midline of the screen, which was either horizontally displaced and stationary or appeared in the center and then moved to the left or right with different velocities. Pursuit responses were investigated to targets moving at different velocities, as well as to positional cues flashed during tracking (Buonocore et al., 2019). To estimate the position-related influence on pursuit, participants were tracking a moving target, and changes in eye velocity were observed, dependent on whether a positional cue was flashed either in front or behind the target movement. Leveraging individual differences in oculomotor behavior across different tasks allowed us to assess the relevance of the performed oculomotor behavior (saccade or pursuit) and the relevant sensory information (position- or velocity-related).

Figure 1
Overview of Tasks for Saccadic and Pursuit Eye Movements



Note. Please note that the dimensions, contrast of the stimuli, and representation of the eye position (gray eye) were adjusted for illustrative purposes. Experiments were conducted with a low-contrast Gaussian blob in front of a gray background, and observers were supposed to look at the targets. Each observer completed all tasks. The experiments were separated based on the relevant sensory information (either position or velocity) and the type of task (saccade, pursuit). In short, observers needed to either make comparable saccades to static (Task 1) or moving targets (Task 2), pursue a target where additional position cues were flashed (Task 3), or just pursue a moving target (Task 4). For more details, see the text and the Method section.

An illustration of the results of the individual tasks can be seen in Figure 2. The top row shows the average saccade endpoints for each observer relative to the target position. Saccades to static (Figure 2A) and moving targets (Figure 2B) were quite accurate. Average errors were usually below 1 degree of visual angle [deg], indicating a successful integration of position- and velocity-related information for saccadic eye movements. Nevertheless, there is substantial variability in the landing positions across observers. Since the initial vertical position step was identical for saccades to stationary and moving targets, we extracted the horizontal saccade target error as a reflection of the performance for saccades to stationary or moving targets. Note that we compute the relative error, where negative values mean consistent undershoots and positive values are consistent overshoots of the target. In the pursuit–position task, we found that the eye velocity was modulated by the position of the cue. This finding aligns with previous results (Buonocore et al., 2019). Consistent with their original article, when we compared the condition with flashes to the baseline without a flash, we noticed a general decrease in the eye velocity due to the distractive effect of the flash. However, this effect was modulated by the position of the cue: pursuit was faster when the cue appeared ahead and slower when it was behind the target (Figure 2C). This indicates that the pursuit response also compensates for positional errors. To estimate the positional influence on pursuit eye movements, we computed the difference in eye velocity between the condition where the positional cue was in the front versus back (see the Method section for more details). For the pursuit–velocity task, observers had to track target movements with different velocities—the average eye velocity in response to these target movements is shown in Figure 2D. As a measurement for pursuit performance, we extracted the average pursuit gain across the different velocities.

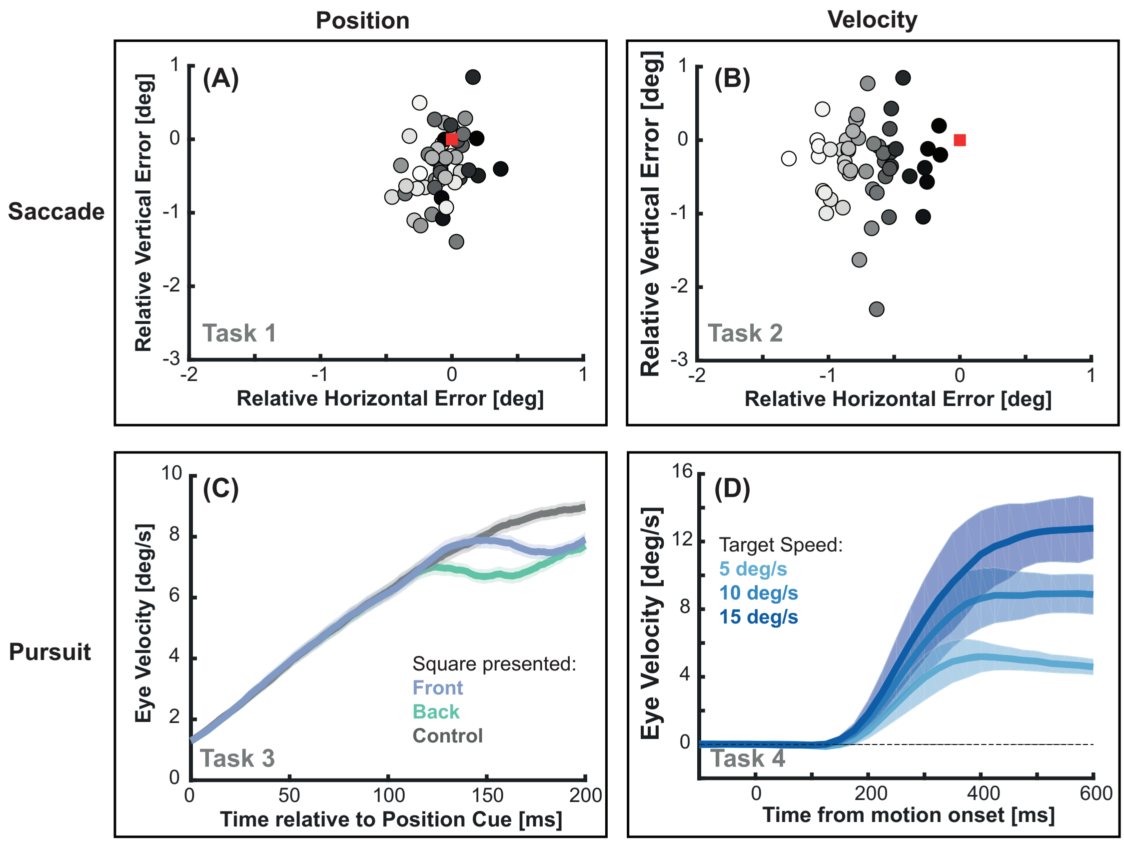
Relevant Sensory Information Links Behavior Across Different Tasks

Based on the individual differences in eye movement behavior, we can now calculate the critical correlations. First, we looked at the relationship between the tasks typically used to study saccadic and pursuit eye movements: saccade performance to a static target and pursuit gain in response to moving targets (Figure 3A). We observed no significant correlation between the behavior in both tasks, $r(47) = -.09$, $p = .523$. This could point to the often assumed separate systems for saccadic and pursuit eye movements. However, due to our balanced combination of tasks, we can also look at the relationships between saccade and pursuit performance when the relevant sensory information is matched. There was a significant correlation between saccadic and pursuit behavior when comparing pursuit gain and saccade error to moving targets, Figure 3B; $r(47) = -.38$, $p = .008$, as well as for the position influence on pursuit and saccade error to stationary targets, Figure 3C; $r(48) = .31$, $p = .026$. This suggests that the absence of a correlation for the typically studied tasks is mostly related to the difference in relevant sensory information. The importance of controlling for sensory information is also supported by the absence of a correlation between the pursuit position influence and saccade accuracy to moving targets, Figure 3D; $r(48) = .04$, $p = .804$.

Together, these results demonstrate that assuming independent and separate systems while only comparing saccades to static targets

Figure 2

Measuring Saccadic and Pursuit Eye Movements With Balanced Sensory Information



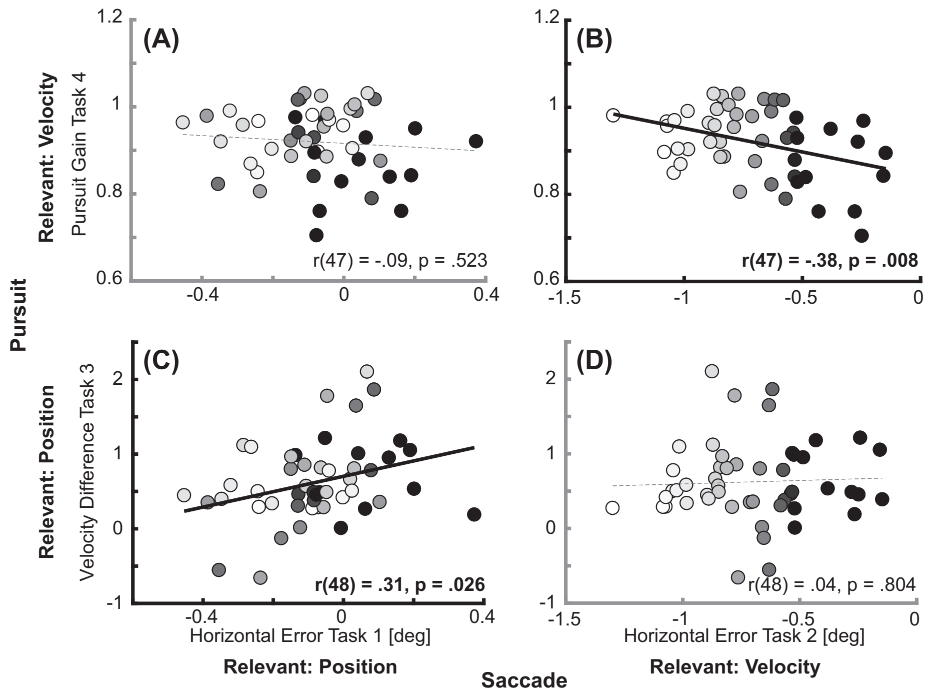
Note. Average relative saccade endpoint for each observer for stationary (A) and moving (B) targets. All endpoints are aligned such that the target is at 0,0 (red square). Dots depict individual observers. Subjects are identifiable by different shades of gray, which are determined based on the horizontal error in Task 2 and stay consistent across all figures. (C) The average eye velocity across observers aligned to the time the additional position cue was presented. The purplish color represents the condition where the cue was presented in front, and the green color is the condition where the cue was presented in the back. The gray condition shows the control condition where no cue was presented. (D) The average eye velocity across observers is shown for different target velocities in the pursuit–velocity experiment. Eye velocity was aligned to motion onset. Please note that panels C and D show the average results across observers; differences between observers can be seen in Figure 3. Shaded areas represent the standard error of the mean.

with pursuit to moving targets is missing critical links in the control of saccadic and pursuit eye movements (Goettker & Gegenfurtner, 2021; Orban de Xivry & Lefèvre, 2007). For velocity information, lower pursuit gain is related to more accurate saccades to moving targets. This could seem counterintuitive at first sight: when there is shared relevant sensory information, one might expect a positive link, for example, higher pursuit gain and more accurate saccades to moving targets. We will address this topic in more detail below when focusing on the coordination of saccadic and pursuit eye movements within the same task. For position information, we observed that observers with a weaker influence of the positional cue on pursuit were also more inclined to undershoot the stationary target with a saccade. In contrast, those with strong pursuit–position influence sometimes even overshot the target position. This phenomenon could potentially be due to a shared gain factor (Tanaka & Lisberger, 2001) for positional information, which simultaneously influences the control of saccadic and pursuit eye movements.

Contributions of Sensory Information and Motor Behavior Across Different Tasks

Our large data set provides an opportunity to estimate the relative contribution of the relevant sensory signal and motor responses across tasks. In simpler terms, we sought to determine whether saccade performance varies depending on the task or whether some observers are generally better at executing saccades. To quantify this, we compared behavior for the same motor responses (saccade or pursuit) either when measured with the same sensory information or when measured with different sensory information (see Figure 4 and the Method section for more details). By examining the differences in the magnitudes of these correlations, we were able to estimate the importance of the relevant sensory information. For example, if the sensory information were irrelevant and the oculomotor response of greatest importance (e.g., the ability to make saccades), we would expect correlations of similar magnitudes regardless of the relevant sensory information.

Figure 3
Link Across Eye Movements



Note. Four correlations are shown, grouped based on the respective oculomotor response (saccade vs. pursuit) and relevant sensory information (position vs. velocity). The metrics selected are pursuit gain in Task 4, the difference in velocity when the positional cue was presented in front or back in Task 3, the horizontal saccade error for targets moving horizontally (Task 2), and the horizontal saccade error for static targets (Task 1). Each point represents one observer, and significant correlations are indicated with a solid black regression line. The brightness of the dots is consistent with the horizontal error in Task 2 and stays consistent across all panels.

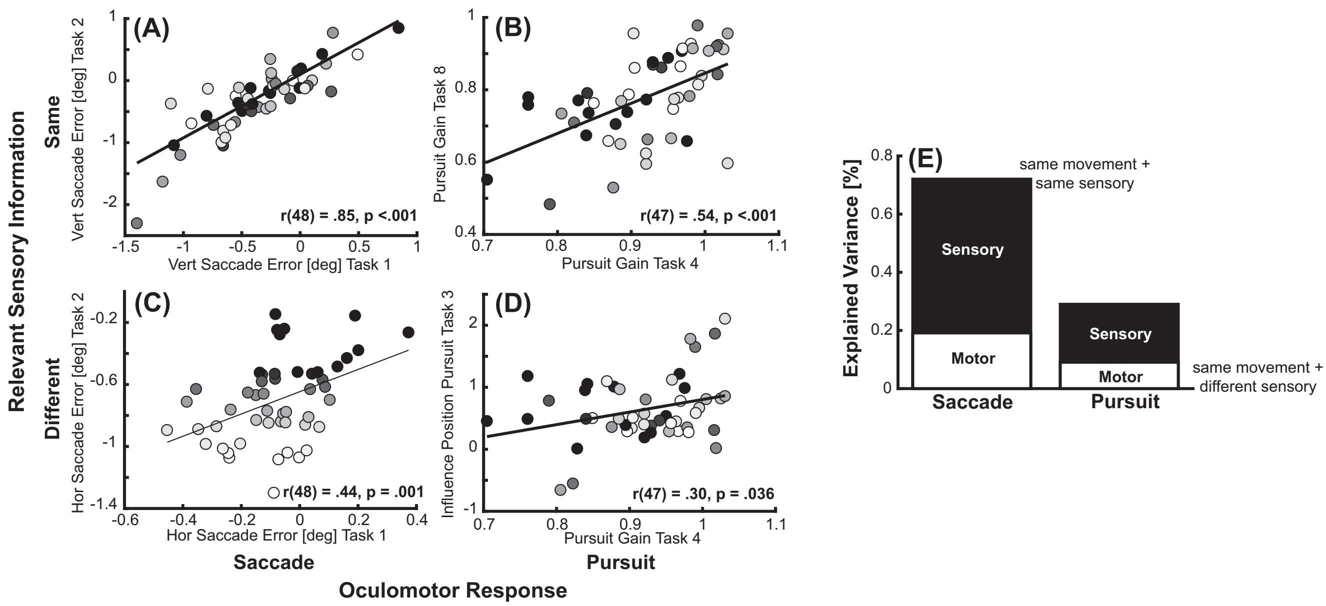
For both eye movements, we observed high correlations when the relevant metric was related to the same sensory information, Figure 4A and 4B; saccade: $r(48) = .85$, $p < .001$; pursuit: $r(47) = .54$, $p < .001$. This replicates that individual differences in oculomotor behavior across comparable tasks are highly reliable. For the critical comparison, we now correlated performance for the same oculomotor behavior, but now when different sensory information is relevant. In this case, for both eye movements, the magnitude of the correlations was only about half the size, Figure 4C and 4D; saccade: $r(48) = .44$, $p = .001$; pursuit: $r(47) = .30$, $p = .036$. To estimate the contribution of the relevant sensory information for eye movement performance across tasks, we computed the proportion of the explained variance when different sensory information was used in comparison to the explained variance in the baseline with the same sensory information (see Figure 4E). Sensory information explained roughly three quarters of the total explainable variance (saccade: 73%, pursuit: 69%), leaving one quarter of explainable variance for the oculomotor response.

We validated our assumption of the critical role of sensory information using a data-driven approach. We performed a principal component analysis (PCA) on the extracted metrics for the isolated eye movement measures (see the Method section for more details), which allowed us to find underlying factors that can explain variance across the metrics. If one assumes that saccadic and pursuit eye movements are based on independent neural control systems, one could expect that the metrics cluster based on the respective oculomotor behavior. In contrast, based on the previously observed correlations, we expect that the underlying factors mainly cluster along the relevant sensory information. We observed that, across all 10 metrics extracted for the isolated eye movement tasks, four factors obtained an Eigenvalue above 1 and could together explain 75% of the variance (see Table 1). The first factor was related to eye movement performance for moving targets (pursuit gain, pursuit latency, and horizontal saccade error to moving targets), combining saccadic and pursuit metrics for the same sensory information. This pattern repeated for the other factors: The second factor was specifically related to saccadic eye movements to stationary targets, whereas the third factor again represented a mixture of saccadic and pursuit metrics related to positional cues with the horizontal saccade error to a position target, the position influence on pursuit, as well as pursuit acceleration. The fourth factor was related to saccade latency.

Together, these results converge on the fact that variations in oculomotor behavior across tasks are mainly driven by the relevant sensory information. We did not observe evidence for the often-present assumption of separate neural control systems of saccadic and pursuit eye movements. Such a distinction would have shown in high correlations between tasks where the same eye movement

Figure 4

Sensory Versus Motor Variability



Note. (A–D) Correlations for pursuit and saccadic eye movements when either the same or different sensory information was relevant. The metrics used are pursuit gain for Task 4 and Task 8, the positional influence on pursuit measured in Task 3, vertical saccade error for the stationary targets (Task 1) or horizontally moving targets (Task 2), and horizontal error to stationary (Task 1) and horizontally moving targets (Task 2). The presentation of the data is similar to that of the top panel. Each point is one observer, with significant correlations indicated by a solid black regression line. As in other figures, the brightness is determined by the horizontal error in Task 2. (E) The sources of explained variance are estimated by the different correlations. The height of the bar shows the explained variance by the same oculomotor response and the same sensory information. The white part of the bar is the explained variance by the same oculomotor response with different sensory information correlations. Therefore, the difference should be a coarse estimate of the contribution of relevant sensory information.

was measured independently of the respective task. Instead, we observed that saccadic and pursuit behavior was correlated with matched sensory information and that the relevant sensory information modulated the strength of the relationship between different saccadic or pursuit measurements across tasks. This suggests that a high-level distinction across tasks should not be based on the executed eye movements but should be based on the relevant sensory information.

Table 1

Underlying Dimensions of Metrics Extracted for Tasks 1–4

Metric	PC1	PC2	PC3	PC4
Pursuit gain	**−0.861**	−0.041	0.231	−0.137
Pursuit latency	**0.814**	−0.136	−0.098	−0.049
Saccade HorError velocity task	**0.604**	−0.012	0.368	−0.014
Saccade VerError velocity task	−0.056	**0.972**	−0.037	0.083
Saccade VerError position task	−0.016	**0.940**	−0.019	−0.133
Pursuit position influence	−0.395	−0.021	**0.814**	0.189
Pursuit acceleration	−0.045	−0.155	**0.792**	−0.172
Saccade HorError position task	0.247	0.238	**0.624**	0.025
Saccade latency velocity task	0.011	−0.108	0.030	**0.885**
Saccade latency position task	0.107	0.061	0.004	**0.898**
Eigenvalue	2.347	2.085	1.658	1.394
Proportion explained variance	0.235	0.209	0.166	0.139
Cumulative variance	0.235	0.443	0.609	0.748

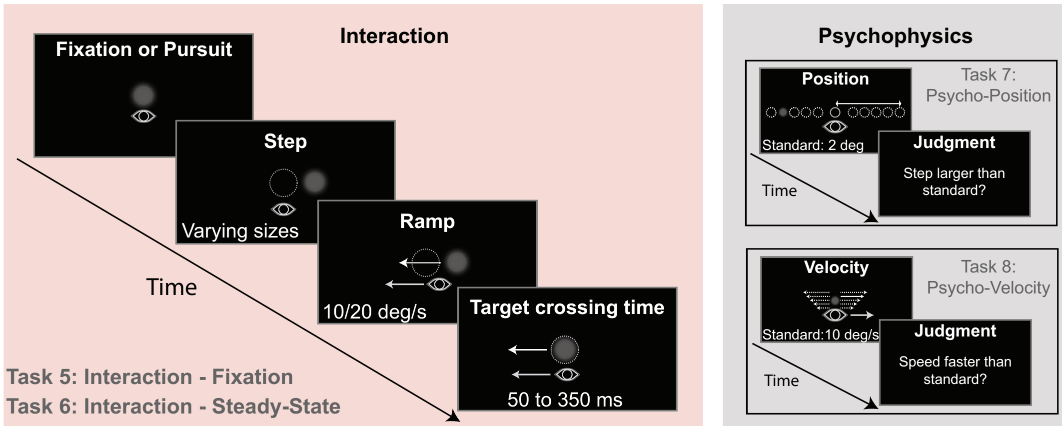
Note. Factor loadings for metrics extracted for the isolated eye movement experiments on the first four principal components (PC). Shading reflects the strength of the evidence. Light Gray, weights above 0.2, darker gray with bold font dominant variables. HorError and VerError are abbreviations for horizontal and vertical errors.

Variability in the Coordination of Saccadic and Pursuit Eye Movements

Across different tasks, variations in isolated saccadic and pursuit eye movements revealed the importance of the relevant sensory information. However, saccadic and pursuit eye movements often also occur together within the same tasks. Therefore, we asked how saccadic and pursuit eye movements are coordinated when the same sensory information is relevant. For this, we studied the interaction and coordination of saccadic and pursuit eye movements when tracking moving targets.

The step ramp paradigm, introduced by Rashbass (1961), is the classic paradigm for studying pursuit eye movements. In this paradigm, the target initially steps in one direction and then immediately starts to move in the opposite direction (see Figure 5). A particular combination of step size and target speed can lead to a tracking of the target with pure pursuit (Figure 6B), while other combinations often require additional corrective saccades (Figure 6A and 6C). A critical factor for whether corrective saccades are needed is the target crossing time. This is defined as the ratio of the initial position step and the target speed. It specifies when the target crosses the initial fixation location (de Brouwer et al., 2002;

Figure 5
Tasks for Measuring Saccade–Pursuit Interactions and Psychophysics



Note. *Interaction:* Depending on the task, observers were fixating or already pursuing a moving target (in this case, there was already an initial target step with a following movement, not illustrated here). In both paradigms, the critical timepoint was the target step with the following change in target velocity (either from 0 to 10 deg/s for fixation or from 10 to 20 deg/s for pursuit). By changing the target step, the time until the target crossed the position it had before the target step varied between 50 to 350 ms of target crossing time. *Psychophysics:* psychophysical tasks for position- or velocity-related information. Observers needed to make perceptual judgments about the size of a target step or the speed of a moving target and compare it to a memorized standard velocity.

Gellman & Carl, 1991). In our study, we systematically varied the target crossing time (see the Method section for more details) and analyzed the probability of corrective saccades (Figure 6D). We made these measurements both when the eyes were initially fixating and during steady-state pursuit (Tasks 5 and 6), considering the potentially faster pursuit dynamics and different relevant information while already pursuing a target (Tavassoli & Ringach, 2009; Wilmer & Nakayama, 2007).

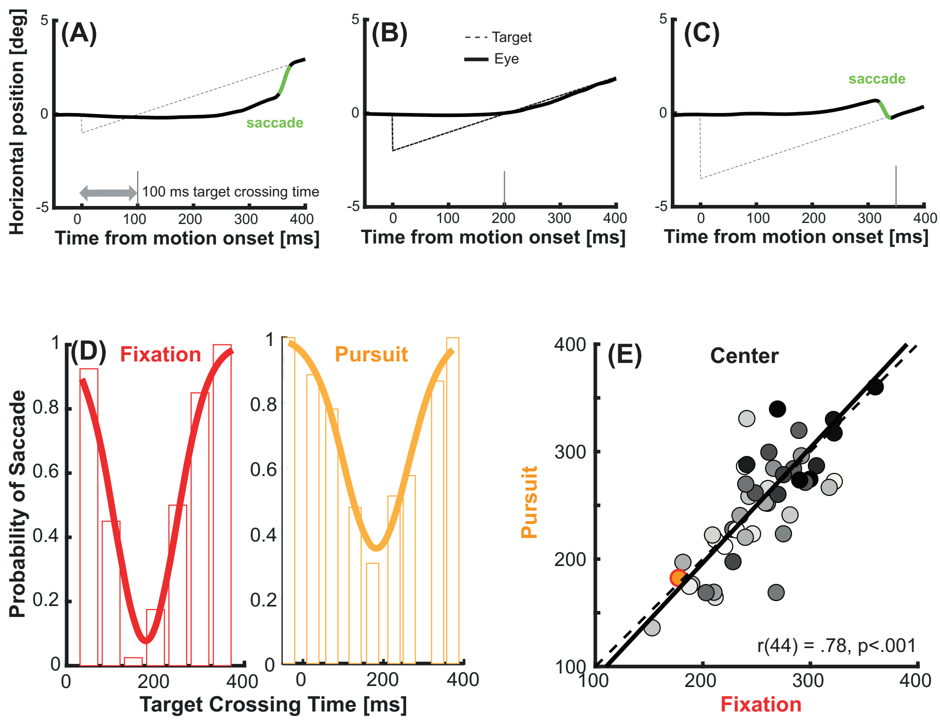
Using the probabilities of corrective saccades across different target crossing times, we fitted an inverse Gaussian to the data (see the Method section for details). This allowed us to quantify saccade–pursuit interactions as a "smooth zone" for each observer (see Figure 6D). Note that four observers were excluded from the analysis due to unreliable fits. The "smooth zone" represents the range of target crossing times where the probability of corrective saccades is decreased (Goettker et al., 2018). The most interesting parameter is the center of the smooth zone. It indicates the combination of position and velocity errors that led to the minimum number of corrective saccades. For both tasks, the smooth zone fit was a good descriptor of the variations in behavior (fixation: average R^2: .94 ± .06; steady-state: R^2: .91 ± .12).

Similar to the investigation of isolated eye movements, we found substantial variability in the center of the smooth zone across observers (see Figure 6E). It ranged between target crossing times of 150–400 ms across observers. These differences reflect that the same combination of errors could be tracked exclusively with pursuit by some observers, while other observers would mainly exhibit pursuit with additional corrective saccades. We observed that the center of the smooth zone was comparable when measured for fixation or during steady-state, $t(45) = 0.92$, $p = .362$, and the similarity was accompanied by a high correlation between the two estimates, with $r(44) = .78$, $p \leq .001$. This suggests similar mechanisms for triggering corrective saccades during fixation and pursuit (see Badler et al., 2019), which combine position- and velocity-related signals independent of potential differences in dynamics (Tavassoli & Ringach, 2009; Wilmer & Nakayama, 2007). However, there were also some significant differences. The probability of saccades at the center of the smooth zone was higher during steady-state pursuit ($M = 0.21$ for fixation versus $M = 0.29$ for steady state), $t(45) = 3.49$, $p = .001$, and the smooth zone got wider during steady-state ($M = 88.65$ ms for fixation versus $M = 102.98$ ms for steady state), $t(45) = 2.66$, $p = .011$. Together, these results can be explained in the context of a recent model (Coutinho et al., 2021): A noisier sensory signal during pursuit (Freeman et al., 2010) in combination with a fixed threshold for triggering saccades leads to a larger variability in target crossing times that elicit a corrective saccade (Nachmani et al., 2020). This matches the observed increase of the minimum and width of the smooth zone during steady-state pursuit.

Tracking Behavior Is Tailored to the Respective Strengths of the Observer

We observed substantial variability in how observers coordinate saccadic and pursuit eye movements. But where are these differences coming from? One possibility is that these differences are not mere noise, but instead reflect meaningful individual strategies to optimize tracking performance. As previously noted, there is a similar level of diversity in the performance of isolated saccadic and pursuit eye movements across observers. This suggests that observers might incorporate these differences when coordinating the two types of eye movements within the same task. To test this, we calculated correlations between the performance in other

Figure 6
Saccade–Pursuit Interactions



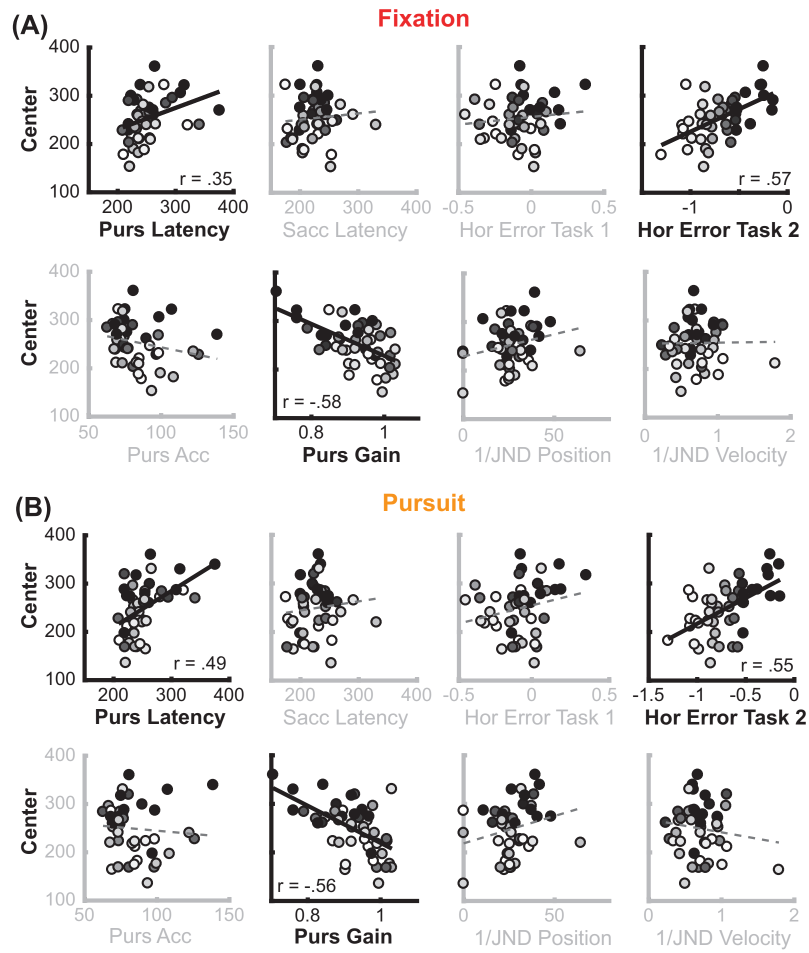
Note. (A–C) Examples of trials in the fixation condition (Task 5) with different target crossing times. Plotted is the horizontal target (dashed) and horizontal eye (solid line) position over time. Target crossing time increases from left to right (from 100 to 350 ms). Please note that only in the left and right panels is there an additional corrective saccade (green line) in the initial following response, whereas no saccade is present in the example trial with a target crossing time of 200 ms. (D) Examples showing the "smooth zone"—our measurement of saccade pursuit interactions for fixation (left) and steady-state pursuit (right). Bars show the probability of a corrective saccade during the initial following response across different target crossing times. Please note here that the analysis was performed based on corrected target crossing times to account for the consistent lag behind the target in the pursuit condition. We fitted an inverse Gaussian to the data and extracted three parameters: the minimum, the mean (center of the function), as well as the width of the zone (see the Method section for more details). (E) Comparison of the center of the smooth zone between fixation and steady-state pursuit. There is large variability in the center. The colored dot represents the data for the representative observer shown in the bottom left; the brightness of the dots for the other observers is fixed across all figures.

oculomotor tasks (Tasks 1–4) and the center of the smooth zone. We did this separately for the fixation as well as the steady-state pursuit condition. Furthermore, to account for potential differences in sensory information quality, we ran two additional psychophysical tasks (Tasks 7 and 8, see the Method section for details). We then examined how these tasks correlated with the center of the smooth zone.

For the fixation condition (Figure 7A), we observed significant correlations between the center of the smooth zone and the horizontal saccade error for the velocity task, $r(44) = .57$, $p < .001$, and pursuit gain, $r(44) = -.58$, $p < .001$. Pursuit latency, $r(44) = .35$, $p = .02$, did show a significant correlation, but this did not survive the correction for multiple comparisons (threshold was $p = .004$ due to 12 correlations tested). For the links to the smooth zone in the steady-state pursuit condition (Figure 7B), we observed the same pattern of results: The center of the zone correlated significantly with pursuit latency, $r(44) = .49$, $p < .001$, the horizontal saccade error for the velocity task, $r(44) = .55$, $p < .001$, and pursuit gain, $r(44) = -.56$, $p < .001$. The other oculomotor metrics and the psychophysical sensitivities measured for discriminating position steps and different target velocities did not show a significant correlation. These results suggest that observers with a better pursuit (low latency and high gain) use it more to catch up with moving targets moving away from the eyes (reflected in an earlier center). Observers with higher saccade accuracy for moving targets rely more on those to catch up with moving targets (reflected in a later center).

But do these correlations demonstrate a meaningful strategy to optimize tracking behavior given the strengths of an observer? One could argue that these correlations mainly emerge due to variations in pursuit behavior. For observers with bad pursuit (long latency and low gain), the eyes move slower, and therefore, the center of the smooth zone, corresponding to the stimulus with the least need for additional corrective saccades, should occur later. However, the

Figure 7
Link Between Metrics and Smooth Zone

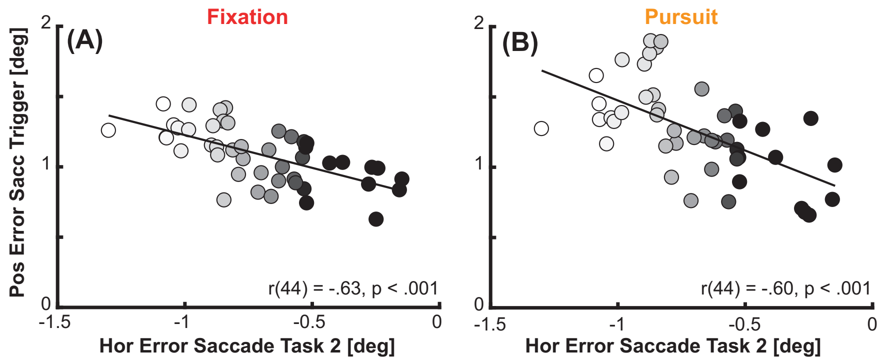


Note. (A) Shown are correlations between the center of the smooth zone for the fixation experiment estimated based on corrected target crossing times and different metrics extracted from the other experiments. Each data point reflects the data from one observer. The shading of data points is consistent across all figures. Lines reflect a linear regression fitted to the data; the line is solid if the correlation reaches significance. (B) Same as on the top, but this time with the center of the smooth zone for the steady-state pursuit experiment.

correlation with saccade accuracy to moving targets suggests a more specific strategy: First, the relation between the accuracy of the saccades to moving targets and the center of the smooth zone was independent of variations in pursuit behavior since partial correlations controlling for differences in pursuit latency and gain were also significant, fixation: $r(44) = .47$, $p = .001$; steady-state: $r(44) = .46$, $p = .002$. Second, looking specifically when the corrective saccades were triggered, we found a strong correlation across all trials between the average position error that was present at the moment these corrective saccades were presumably planned (125 ms before saccade execution) and the saccade accuracy of moving targets, see Figure 8; fixation: $r(44) = -.63$, $p < .001$; steady-state: $r(44) = -.60$, $p < .001$. This suggests that people with accurate saccades to moving targets triggered corrective saccades already with smaller position errors, and this relation also remained significant when controlling for the location of the center of the smooth zone, fixation: $r(44) = -.40$, $p = .007$; steady-state: $r(44) = -.49$, $p < .001$. Third, the correlation can also not be explained by overall differences in oculomotor performance. There was a negative correlation between saccadic accuracy and pursuit gain for moving targets (see Figure 3), and the center of the smooth zone was not correlated with the accuracy of saccades to static targets. Therefore, these results cannot be interpreted as observers with generally good and accurate eye movements (e.g., high pursuit gain and high saccadic accuracy) having earlier centers of the smooth zone.

Together, these results suggest that within the same task, observers tailored the coordination of saccadic and pursuit eye movements to their respective strengths. On the one hand, participants with good pursuit performance (lower pursuit latency and higher pursuit gain) showed an earlier center of the smooth zone. Thus, they relied on pursuit to quickly catch up to the target without additional corrective saccades. On the other hand, observers

Figure 8
Saccade Trigger and Accuracy to Moving Targets



Note. Shown are correlations between the accuracy of saccades to moving targets (measured as horizontal error in Task 2) and the average position error that was present 125 ms before the saccade was initiated. (A) Shows the average position error for Task 5, (B) The average position error for Task 6. The coloring of the dots is similar to the other figures.

with accurate corrective saccades to moving targets showed later centers of the smooth zone, thus they relied on them more frequently and triggered them with less position error to catch up with targets moving away from the eye. Conversely, they use them less frequently when the target needs more time to cross the fixation (long target crossing times). These results speak for a fine-tuned specialization in the coordination of saccadic and pursuit eye movements within a tracking task.

Discussion

Due to their importance for visual perception, eye movements are often considered to be a "window into the mind" (König et al., 2016; Schütz et al., 2011; Shaikh & Zee, 2018; Thakkar et al., 2017; van Gompel et al., 2007). They offer a unique advantage as objective, implicit, and continuous indicators of sensory processing (Lisberger, 2015) and decision-making (Spering, 2022), underpinned by well-understood neurophysiological mechanisms (Leigh & Zee, 2015). However, traditional studies tend to focus on average behavior within a single type of eye movement and a specific task. Our findings demonstrate that this approach misses out on critical links and variations in the control of oculomotor behavior. When comparing saccadic and pursuit eye movements across different tasks, we could show that individual variability across different tasks is mainly related to the relevant sensory information (position vs. velocity) rather than the relevant eye movement (saccade vs. pursuit). This finding calls into question the traditional distinction of different processing streams for saccadic and pursuit eye movements (Klein & Ettinger, 2019; Leigh & Zee, 2015; Robinson, 1986). Our results have profound implications for the theoretical understanding of the sensorimotor processing hierarchy. They necessitate a reevaluation of previous data used to map brain circuits for oculomotor control. Additionally, they underscore the importance of moving beyond average observations to embrace individual differences as a rich source of information. These individual differences not only reveal the strengths and weaknesses of observers. When combined across various tasks, they allow insights into *why* observers behave differently in a given task.

By now, multiple studies have shown strong interactions and shared information between saccadic and pursuit eye movements at the behavioral (Goettker & Gegenfurtner, 2021; Orban de Xivry & Lefèvre, 2007) and neurophysiological levels (Krauzlis, 2004, 2005). Our unique data set allowed us to take this line of work one step further by using individual differences to investigate relationships between different oculomotor tasks. A critical assumption behind that approach is that these differences across observers are systematic and reliable. Previous research has shown that individual differences in saccadic and pursuit eye movements have a good retest reliability (Bargary et al., 2017; Ettinger et al., 2003; Versino et al., 1993), are stable for fixation preferences in images (Castelhano & Henderson, 2008; de Haas et al., 2019) or effects of a structured background (Schröder et al., 2021). The stable correlations between tasks confirm and extend previous work (Calkins et al., 2003; Iacono & Lykken, 1981), strongly suggesting that variations of oculomotor behavior reflect a trait of the observer (Bargary et al., 2017; Lohr et al., 2022; Rigas et al., 2016).

The study by Bargary et al. (2017) measured a set of different oculomotor tasks across observers and suggested that a major contributor to oculomotor behavior might be related to the type of task and not necessarily to the eye movements themselves. We extend their work by adding the crucial conditions of saccades to moving targets and pursuit in response to position-related cues. This unique combination of tasks allowed us to firmly link the individual differences across these tasks. Saccadic and pursuit performance are uncorrelated when the relevant sensory information is different. However, when relevant sensory information is matched, the performance of saccadic and pursuit eye movements is correlated (Figure 3). A data-driven approach supported the separation of tasks based on the relevant sensory information (see Table 1).

In most previous studies, the paradigms used for saccadic or pursuit eye movements differ fundamentally with respect to the relevant sensory information (position-related information for saccades and velocity-related information for pursuit). Neuroimaging studies often have used these comparisons to dissociate areas in the brain responsive to saccadic or pursuit eye movements (Agtzidis et al., 2020; Lencer & Trillenberg, 2008; O'Driscoll et al., 1998; Petit et al., 1997;

Rosano et al., 2002). Here, we provide direct behavioral evidence that this confound could lead to unintentionally specious interpretations since these studies cannot distinguish between a motor-related response (saccade vs. pursuit) and a sensory-related response (position vs. velocity). In neurophysiology, some studies already tried to tackle this problem. For example, the superior colliculus, which is mainly thought to be related to saccade control (Gandhi & Katnani, 2011; Sparks, 1986), seems to provide a more general position error signal, which is also relevant for pursuit eye movements (Basso et al., 2000; Krauzlis et al., 2000). Moreover, the middle temporal area, mainly thought to be involved in pursuit control (Dubner & Zeki, 1971; Pack & Born, 2001), seems to provide a more general velocity-related signal: Lesioning this area affects not only the pursuit control but also the saccadic eye movements to moving targets. Saccades to stationary targets stay intact (Newsome et al., 1985; Newsome & Paré, 1988). Even the well-known omnipause neurons in the brain stem (Sparks, 2002), which are thought to be a gating mechanism for saccadic eye movements, are involved in the control of pursuit eye movements (Missal & Keller, 2002). Therefore, there seems to be converging evidence on the neurophysiological level that responses in single brain regions are not always related to the control of a specific type of eye movement (Sun et al., 2017, 2022). We emphasize that the assumption that there are still some areas selectively related to the control of saccadic and pursuit eye movements is valid. Not every region initially identified as a region for a specific type of eye movement has to be exclusively related to sensory information. However, due to the potential confound in the relevant sensory information, the interpretability of such comparisons is often limited. Future studies should make use of additional control conditions, for example, by measuring saccades to moving targets or by comparing variations in the combination of saccadic and pursuit eye movements due to trial-by-trial variability (Goettker, Brenner, et al., 2019; Goettker et al., 2018) to be able to directly dissociate eye movement-related from sensory-related areas.

When examining the coordination of saccadic and pursuit eye movements within the same task and type of sensory information, we noticed that observers adjusted their behavior to their individual strengths (Figure 8). Interestingly, this coordination was related more closely to the performance in the isolated oculomotor responses than to differences in the sensory quality. While previous studies have observed links between individual differences in sensory processing and isolated eye movement behavior (Wilmer & Nakayama, 2007), the coordination of different types of eye movements presents a more complex scenario. For optimal tracking performance, a strategic adjustment should be related to the performance of the isolated eye movements, which are based on a combination of different sources of sensory and motor noise. Therefore, measurements of the sensory noise alone might not be sufficient to explain the coordination of different types of eye movements. Regarding the variations in isolated oculomotor behavior, we found a negative correlation between the performance of pursuit and that of saccades to moving targets. This finding rules out the idea that differences in the coordination of saccadic and pursuit eye movements are merely a result of overall superior oculomotor performance (saccade and pursuit eye movements were not both more accurate across observers). It also dismisses the hypothesis of a generally improved accuracy for one type of eye movement (e.g., accuracy of saccades to stationary targets was not related to the interaction). Instead, it suggests a specific behavioral optimization within a given task based on the performance of the individual eye movements under those particular conditions.

With the previously established importance of shared sensory information that links saccade and pursuit control across tasks, this leaves the puzzling question of why saccadic and pursuit eye movements do not show comparable performance when tracking a moving target. We believe that this reflects a hierarchical organization in the control of oculomotor behavior. There is an underlying distinction in the processing of position- and velocity-related information (Goettker, Braun, et al., 2019), which can explain the links in performance across tasks, but for a specific task with the same sensory information, there is a further specialization in the control of individual eye movements.

So, how could the performance of saccadic and pursuit eye movements differ despite shared sensory information? The origin of variability in eye movement responses has been debated extensively for eye movements (van Beers, 2007). While some reports concluded that variability was, to a large extent, based on sensory information alone (Osborne et al., 2005, 2007), other studies reported an important role of motor and decision noise (Egger & Lisberger, 2022; Gegenfurtner et al., 2003; Rasche & Gegenfurtner, 2009). Either way, any form of additional motor noise, central decision noise, or differences in how the sensory signals are transformed into the respective eye movement could explain a negative correlation between saccadic and pursuit eye movement performance within the same task despite shared sensory information. The reasons behind such specialization could be manifold and, unfortunately, cannot be addressed with our correlational data alone. On the one hand, observers could have learned that one type of their eye movements is more accurate when tracking moving targets, so they rely on that one more frequently. On the other hand, if observers, for example, initially relied more on corrective saccade when tracking moving targets, the accuracy of these movements could just have improved with practice.

It is discussed elsewhere how position- and velocity-related sensory errors are integrated into decisions on whether to perform corrective saccades (Coutinho et al., 2021; de Brouwer et al., 2002; Gellman & Carl, 1991; Goettker & Gegenfurtner, 2021; Nachmani et al., 2020). Our results now extend these findings across the single trial level and show that the decision criteria differ widely across observers. They are probably based on a life-long learning process and optimized for each individual's tracking performance. The fact that variability in behavior is more than just noise in measurements but can be related to the strengths of the particular observer has already been suggested for the relation between eye movements and face perception. Although there is a theoretical optimum for where to look at a face, each observer has their own preferred location for where to fixate, and this correlates with their face identification performance (Peterson & Eckstein, 2013). In addition, people who spend more time looking at faces are also better at recognizing faces (de Haas et al., 2019). Related to that, a recent study showed that superrecognizers (Ramon et al., 2019), people with exceptional abilities to recognize faces, spend significantly more time looking at faces than controls and also look closer at a theoretical optimal location for face identification (Linka et al., 2022). Thus, individual differences might, in general, reflect the strengths and weaknesses of sensory processing and motor behavior for each observer.

Our results provide direct empirical evidence that saccade and pursuit performance are correlated across different tasks, but only when tested with matched sensory information. This suggests that the high-level dissociation often assumed to occur between the control of saccadic and pursuit eye movements is instead related to the difference in the relevant sensory information (see Figure 9). Both types of eye movements are controlled by a combination of independent position- and velocity-related information (Goettker, Braun, et al., 2019). This is quite similar to the control of other kinds of actions, for example, interception (de la Malla et al., 2018; Smeets & Brenner, 1995). When coordinating saccadic and pursuit eye movements, differences across observers revealed that they adjust their tracking behavior according to their strengths (Peterson & Eckstein, 2013). These new insights make it very clear that future studies addressing the underlying brain areas and mechanisms for motor control need careful control for the relevant sensory stimuli and can provide exciting pathways toward a deeper understanding of the underlying neurophysiological circuits. Furthermore, the differentiation of relevant sensory information and executed eye movement might provide new insights for clinical research, for example, by demonstrating that patient groups have no deficits related to a certain type of movement, but to the sensory processing relevant for the characteristics of the task (Fooken et al., 2022).

Method

Experimental Design

The goal of this study was to measure individual differences across a set of different oculomotor and psychophysical tasks and then look at their relations. We were specifically interested in the role of the relevant sensory information for oculomotor control across different tasks and how observers coordinated their saccadic and pursuit eye movements within a task. The number of participants, study design, hypotheses, and parts of the analysis were preregistered at https://doi.org/10.23668/psycharchives.12502 (Goettker, 2021).

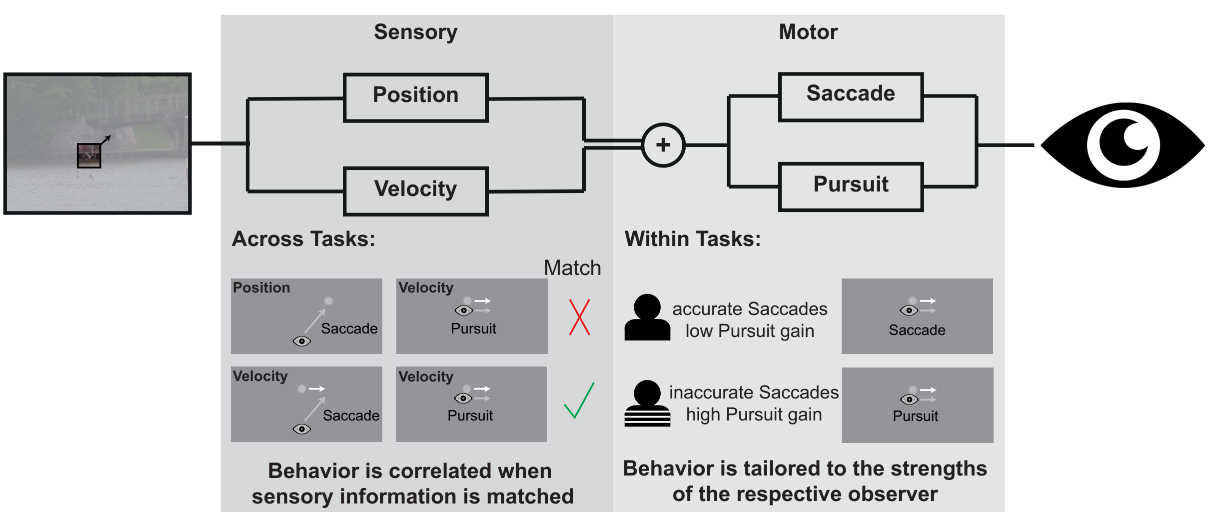
Observer

A total of 50 observers (M_{age}: 24.42, std = 3.86, range: 19–35) took part in the study; 37 identified as female, 13 as male. All observers were naïve concerning the purpose of the study and had normal or corrected-to-normal vision. Before the start of the experiment, they gave informed consent, and all tasks were approved by the local ethics committee (Giessen University Local Ethics Commsion 2017-0029) and were conducted in accordance with those guidelines at the Leibniz Institute for Psychology (ZPID) in Trier. Observers were compensated with money for their efforts during the experiment.

Setup and Experimental Conditions

Observers were seated at a table in a dimly illuminated room with their heads positioned on a chin rest. In this position, their eyes were roughly aligned with the height of the center of a monitor (53 cm × 30 cm, 1,920 × 1,080 pixels, 144 Hz, BenQ, Taipei, Taiwan) with a viewing distance of 90 cm. Under these circumstances, the monitor spanned approximately 34 × 19 degrees of visual angle. The tasks were programmed and controlled with MATLAB 2020a (MathWorks, Natick, Massachusetts) using Psychtoolbox (Kleiner et al., 2007). Gaze was recorded from one eye with a desk-mounted eye tracker (EyeLink 1000 Plus, SR Research, Kanata, Ontario, Canada) at a sampling frequency of 1,000 Hz. To ensure accurate recordings

Figure 9
Integrative Oculomotor Framework



Note. The velocity and position of a potential eye movement target are analyzed. Both information streams are then used to trigger a combination of saccadic and pursuit eye movements to bring the eye toward the target. Across different tasks, saccadic and pursuit eye movement behavior is correlated when the sensory information is matched (e.g., for a saccade to a moving target and pursuit of a moving object, but not for a saccade to a stationary target and pursuit of a moving object). Within the same task, the decision when to trigger a saccadic or pursuit eye movement is based on the individual strengths of a respective observer. An observer with accurate saccades to moving targets is more likely to use a saccade to catch up with a moving target. Another observer with a more accurate pursuit is more likely to use a pure pursuit response for the same trial. See the online article for the color version of this figure.

before each block, a 9-point calibration was performed, and additional drift checks were used at the start of each trial. The drift checks allowed the observers to perform each task in a self-paced manner. To start a trial, they needed to press the space bar when looking at the central fixation.

All observers completed a total of eight different tasks (see Figure 1): (1) saccades to stationary targets, (2) saccades to moving targets, (3) pursuit of moving targets, (4) pursuit with flashed stationary target, (5) smooth-zone measurement out of fixation, (6) smooth-zone measurement out of steady-state pursuit, (7) psychophysical judgment of position steps, and (8) psychophysical judgments of target speed. Tasks 1, 2, 3, 7, and 8 consisted of one block and took about 15 min; Tasks 5, 6, and 7 consisted of two blocks and took approximately 30 min. All tasks were completed across three 1-hr sessions on separate days. Participants completed each task before moving on to the next one, and the order of tasks was randomized for each observer. At the beginning of the first session, additional demographic information was collected. Across all tasks, each observer completed a total of 1,460 trials.

Individual Tasks

Task 1: Saccades to Stationary Targets

The goal of this task was to measure saccade accuracy to static targets. In this and all other tasks, observers were instructed to look at and follow the target as fast and accurately as possible. After the drift check, observers were asked to always look at and then fixate a Gaussian blob target (SD = 0.3 deg, maximum contrast = 0.2) presented in front of a gray background. The target initially appeared in the middle of the screen but shifted downward by 7 deg (position: [0, −7]). After a random time between 1,000 and 1,500 ms (uniformly sampled in steps of 7 ms, which was the duration of one frame given 144 Hz), the target then disappeared and directly reappeared at one of six different positions. It always appeared at the vertical center of the screen but now shifted by either 1.25, 2.5, or 3.75 deg to the left or the right (possible positions: [−3.75, 0], [−2.5, 0], [−1.25, 0], [1.25, 0], [2.5, 0], [3.75, 0]). Due to the starting position, which is the same in Task 2, saccades always had an upward component. These positions were chosen to match the position of the moving target in Task 2 after 250 ms of movement with the different target speeds. The gaze was monitored online to ensure that observers only moved their eyes to the new target after it appeared. If observers moved their eyes too early (represented by vertical eye position higher than −5 deg in the frame before the target was shown), an error message appeared, and the task continued with the next trial. The horizontally shifted target then stayed on the screen for 1,000 ms. The trial ended, and the drift check for the next trial followed. Overall, each observer completed 120 trials (3 Horizontal Positions × 2 Directions × 20 Repetitions).

Task 2: Saccades to Moving Targets

The goal of this task was to measure saccade accuracy to moving targets. The trial structure of Task 2 was identical to Task 1. After an initial fixation on the target on the bottom half of the screen (position: [0, −7]), after a random time between 1,000 and 1,500 ms, the target again jumped to the center. However, here, instead of being horizontally shifted, the target appeared in the center of the screen and immediately moved either to the left or the right. The target could move at 5, 10, or 15 deg/s and disappear after 1,000 ms. Overall, each observer completed 120 trials (3 Speeds × 2 Directions × 20 Repetitions).

Task 3: Pursuit With Flashed Stationary Target

The goal of this task was to measure the influence of positional information on the pursuit system. The design of this paradigm was inspired by a recent article by Buonocore and colleagues (Buonocore et al., 2019). Each trial again started with the Gaussian blob target in the center of the screen for a random time between 1,000 and 1,500 ms. Then the target stepped either 2 deg to the left or 2 deg the right and moved with 10 deg/s toward the center. Just when the target crossed the center of the screen, additional squares could be flashed on the screen for one frame (around 7 ms). The squares were grayish (contrast: 0.1) and 2 by 2 deg large and could either appear at 4 deg left of the center or 4 deg right of the center. Across trials, the squares were therefore either presented in front of the pursuit, behind the pursuit, or as a control and were not shown. Observers were told to ignore the squares and just continue tracking the moving target, which disappeared after 1 s of movement. Observers completed two blocks of 120 trials each (2 Directions × 3 Squares × 20 Repetitions), leading to a total of 240 trials.

Task 4: Pursuit of Moving Targets

The goal of this task was to measure pursuit accuracy to moving targets and is similar to the conventionally used Rashbass paradigm (Rashbass, 1961). Each trial started with a Gaussian blob target in the center of the screen (position: [0,0]) for a random time between 1,000 and 1,500 ms. After this time, the target stepped either to the left or the right and immediately started to back to the center of the screen. The target could move again with either 5, 10, or 15 deg/s, and the size of the step was adjusted so that the target reached the center of the screen after 200 ms with the respective velocity. After crossing the center, the movement continued for 1,000 ms, and the target disappeared. Overall, each observer completed 120 trials (3 Speeds × 2 Directions × 20 Repetitions).

Task 5: Smooth-Zone Measurement out of Fixation

The goal of this task was to assess saccade–pursuit interactions via the smooth zone out of fixation. Comparable to Task 4, it is similar to a Rashbass step ramp paradigm. To assess the smooth zone, we systematically varied the target crossing time. Again, the target stayed in the center of the screen for a random time between 1,000 and 1,500 ms. Then, the target stepped to the left or the right and immediately moved at 10 deg/s toward the center. The time the target took to reach the center varied between 50, 100, 150, 200, 250, 300, or 350 ms by adjusting the size of the initial step accordingly. The target kept moving for 700 ms after it crossed the center of the screen and then disappeared. Each block consisted of 140 trials (2 Directions × 7 Target Crossing Times × 10 Repetitions), and observers completed two blocks for a total of 280 trials.

Task 6: Smooth-Zone Measurement out of Steady-State Pursuit

The goal of this task was to assess saccade–pursuit interactions via the smooth zone out of the steady-state pursuit. For this, we used a movement that included two step ramps and aimed at creating the same target crossing times with comparable position and velocity errors to Task 5 at the second step ramp while observers were already in steady-state pursuit. Trials could move either to the left or right, and this time, depending on the direction of the movement, the target started at 15 deg to the left or right in the opposite direction of the target movement. After a random time between 750 and 1,250 ms, the target started to move with 10 deg/s and the first ramp had a target crossing time of 200 ms (it took 200 ms after the first step to cross over the initial position of the target). Then, the target kept moving toward the center for a random duration between 750 and 1,250 ms. During this time, the eyes were in steady-state pursuit following the target. After the random time, a second step occurred, where the target stepped backward and afterward started moving at 20 deg/s. The size of the step was chosen to match the target crossing times of 50, 100, 150, 200, 250, 300, or 350 ms (by not using the 20 deg/s of the velocity, but 10 deg/s velocity since the eyes were supposed to be already moving at 10 deg/s). In this way, the smooth zone could be compared with comparable target movements during fixation or steady-state pursuit. After the second step, the target again kept moving for 700 ms and then disappeared. Overall, observers completed a total of 280 trials (2 × Blocks × 2 Directions × 7 Target Crossing Times × 10 Repetitions).

Task 7: Psychophysical Judgment of Position Steps

The goal of this task was to measure the perceptual ability to discriminate different target steps. In this task, observers had to judge the size of a target step. Each trial again started with the target in the center of the screen for a random time between 1,000 and 1,500 ms. Afterward, the target could again step either to the left or to the right by 1.2, 1.6, 2, 2.4, or 2.8 deg. After the step the target stayed stationary and visible for a random time between 500 and 700 ms. Then observers had to judge whether the just observed target step was larger than the size of a standard target step. The standard step was 2 deg. To reduce the number of trials, we did not use a two-alternative forced choice task, but a memorized standard, where we presented the labeled standard target 5 times in a row randomly stepping to the left or the right. The standard was shown at the beginning of the task and again before trials 10, 30, 60, and 100. The standard was the average of all visible target steps. The judgements of the different target steps allowed to compute a psychometric curve. Overall, observers completed 150 trials (2 Directions × 5 Step Sizes × 15 Repetitions).

Task 8: Psychophysical Judgment of Target Speed

The goal of this task was to measure the perceptual ability to discriminate different target speeds. The design of the task was identical to Task 7. However, instead of the size of the target step, we varied the velocity of the moving target. The target stayed in the center for a random time between 1,000 and 1,500 ms and then stepped to the left or right and immediately started moving toward the center of the screen. The velocity varied between 8, 9, 10, 11, or 12 deg/s, and the size of the step was again chosen so that the target crossed the center of the screen in 200 ms. These velocities had then to be compared with the standard velocity of 10 deg/s, which again was introduced as a memorized standard. Overall, observers completed 150 trials (2 Directions × 5 Speeds × 15 Repetitions).

Data Analysis and Preprocessing

Eye movement data were analyzed offline using our custom software programmed in MATLAB. Saccades were detected by using the Eyelink criteria of a speed and acceleration threshold of 30 deg/s and 4,000 deg/s^2, respectively. Blinks in the data were linearly interpolated given the next available neighboring samples. Eye positions were filtered using a second-order Butterworth filter with a cutoff frequency of 30 Hz, and afterward, we calculated the horizontal and vertical eye velocity by taking the difference between consecutive samples of the filtered position traces and multiplying it by the sampling frequency to represent it in degrees per second. Eye velocity was then again filtered by a second-order Butterworth filter with a cutoff frequency of 20 Hz. To analyze the data for leftward and right movements or leftward and rightward steps together, we flipped the horizontal positions of the eye movement for targets that moved or stepped to the left. Pursuit onset was detected by a custom algorithm. To calculate pursuit onset, we used a velocity trace where detected saccades were linearly interpolated in the velocity trace (saccade ±30 ms). Then, as a baseline, the velocity and standard deviation of the eye velocity were computed from 25 ms before to 25 ms after the target movement onset. Pursuit onset was the point where the eye velocity was above three times the standard deviation of the baseline and stayed from there for more than 50 ms above 30% of the target speed.

Metrics

For tasks measuring performance of isolated eye movements (Tasks 1–4), we computed individual metrics. If not noted differently, we computed each of the following metrics for each trial and then averaged them across both directions for each of the factor steps (e.g., target step or target velocity). To compare the results across tasks we then averaged across the factor steps to have one value per subject per task. In the saccade tasks (Tasks 1 and 2) we computed the *horizontal and vertical saccade error* (respective difference between eye position and target position at saccade offset) as well as *saccade latency*. The target positions in these tasks were designed in a way that saccades landed at comparable positions, which allowed for a direct comparison. For the pursuit–position task (Task 3) we computed the mean difference between the average eye velocity between 100 and 180 ms after the presentation of the square in front or behind the pursuit eye movement to quantify the influence of positional information on pursuit. We obtained this time window, by looking at the difference in the average eye velocity across subjects for the different conditions (see Figure 2). For the pursuit–velocity task (Task 4) we computed *pursuit latency* based on pursuit onset detection as well as *pursuit acceleration* (defined as the slope of a linear regression fitted to the velocity in a temporal window of 50 around the pursuit onset) and *pursuit gain* (defined as the average velocity from 200 to 300 ms after pursuit onset divided by target speed). Note that for the computation of pursuit gain, saccadic velocity epochs ±30 ms were eliminated from the analysis.

For the tasks measuring the coordination of saccadic and pursuit eye movements by establishing a "smooth zone" (see Figure 5), we labeled each trial as a saccade trial if a saccade occurred within the first 400 ms after target motion onset. If no saccade happened during that interval, the trial was labeled a pursuit trial. We then wanted to compute the average probability of saccade trials across the different presented target crossing times. However, we noticed that, especially in estimating the smooth zone during steady-state, the eyes were not always perfectly on the target, and the position (Average = 1.98 deg for fixation vs. 1.86 deg for steady-state) and velocity errors (Average = −9.82 deg/s for fixation vs. −10.63 deg/s for steady-state) were therefore not directly comparable. To account for that, we performed the analysis using a corrected target crossing time. For that, we computed the horizontal position error between the eye and target position after the relevant target step as well as the velocity error (for the eye, we took the average horizontal eye velocity between 75 ms before and 25 ms after the target step) and calculated the corrected target crossing time based on the negative position error divided by the velocity error. We grouped the computed target crossing times in bins between 0 and 400 ms in 50 ms steps. One additional bin was capturing negative target crossing times between −100 and 0 ms. For bins that had more than three trials, we computed the average probability of a saccade, as well as the median of the computed target crossing time. Based on these values, we then estimated the smooth zone by fitting an inverse Gaussian function, which allowed us to extract three relevant parameters. The mean of the function indicated the *center of the smooth zone* (e.g., the minimum number of saccades at 250 ms target crossing time), the standard deviation of the function reflected the *width of the smooth zone* (e.g. low number of saccades across a width of 80 ms), and the amplitude of function gave us the *minimum of the smooth zone* (e.g., at the center of the smooth zone the probability of a saccade is 0.1). As an additional exploratory analysis, we also looked at what signals were available before the saccade was executed. For that, we computed the average absolute position error 125 ms before saccade onset.

For the psychometric tasks (Tasks 7 and 8), we fitted a cumulative Gaussian as a psychometric function to the average ratings across factor steps. We used the inverse of the just-noticeable differences as an indicator of the *discrimination performance*. For the motion judgment task, we used the same computation as for Task 3 to estimate the pursuit gain per observer in this task.

Statistical Analysis

To assess the structure of the data we had two approaches: The first was looking at a specific combination of correlations to test and visualize which dimension (motor behavior or sensory signals) observers varied. We computed correlations between metrics reflecting saccadic and pursuit behavior in response to position- and velocity-related sensory information. Given our expectation of the link of oculomotor behavior based on the relevant sensory pattern, we expected a specific pattern: Significant correlations for the same sensory information and no significant correlation for the conditions with different sensory information. Given that independent control should have led to no significant correlation, we did not correct for the four correlations here to not favor one hypothesis above the other. To estimate the relative contributions of sensory signals and motor behavior, we used the magnitudes of the correlations for pursuit eye movements based on the same sensory input (pursuit gain in Tasks 3 and 8) and with different sensory input (position influence measured in Task 3 and pursuit gain in Task 4). We then computed the difference between the explained variance of the correlation based on the same sensory input and the explained variance based on the correlation with different sensory inputs. This allowed us to express the relative contribution of sensory information. We took the same approach for saccadic eye movements (same sensory input: vertical error in Tasks 1 and 2; different sensory input: horizontal error in Tasks 1 and 2). Our second approach was purely data-driven, where we used a PCA based on all metrics extracted for the tasks that investigated isolated eye movements. The PCA was conducted in JASP (JASP Team, 2024), with the settings of relevant factors with an eigenvalue above 1. Factors needed to be orthogonal, and we tried to increase the interpretability of the individual factors by using the varimax setting.

For the investigation of the coordination of saccadic and pursuit eye movements, we computed correlations between the center of the smooth zone and the metrics extracted in the tasks focusing on isolated eye movements. Since we computed the correlation between 12 different metrics extracted in Tasks 1–4 and the two psychophysical (Tasks 7 and 8), we corrected the significance level via Bonferroni-correction.

Exclusion Criteria

Single trials were excluded from the analysis if during a single trial, there were more than 500 ms of missing data due to blinks or other reasons. For saccade measurements (Tasks 1 and 2), trials were excluded if no valid saccade was detected that had a latency between 50 and 700 ms and a position error above 5 deg. For pursuit measurements (Tasks 3, 4, and 8), trials were excluded if no valid pursuit onset with a latency between 50 and 700 ms was found or the computed gain following pursuit onset was below 0.3 or above 2. Please note that for Experiment 8, trials not reaching the pursuit criteria were only excluded from the oculomotor analysis but not from the perceptual analysis. Based on these criteria across all tasks we used 70.873 out of 73.000 trials (97%). The proportion of valid trials across experiments was: Task 1 = 92%, Task 2 = 90%, Task 3 = 95%, Task 4 = 95%, Task 5 = 99%, Task 6 = 99% Task 7 = 99%, Task 8 = 98% for psychometric data, 93% for oculomotor behavior.

In addition to the exclusion of single trials, we excluded the results from one observer when analyzing Task 5 due to a high trial exclusion rate (>70%) and an average pursuit gain below 0.5. In addition, we excluded the data from four observers from the analysis of the smooth zone experiments since here, the estimated center of the smooth zone was at least 50 ms outside the range of our tested target crossing times (>400 ms) and, therefore, no reliable estimate was possible.

Deviations From Preregistration

In comparison to the preregistration, we focused the story and analysis presented in the article on two of the three initial hypotheses: the structure of the tasks and the relation of different metrics to the center of the smooth zone. The third hypothesis regarding the comparison of the smooth zone between fixation and steady-state pursuit is still included and discussed. With respect to the analysis, we needed to exclude a few participants from parts of the analysis (see exclusion criteria). With respect to the analysis, we slightly adjusted our initial plan. With such a large data set, there are a lot of potentially interesting results and possible comparisons, and in the preregistration, we mentioned a lot of different metrics to compute. To keep the analysis more focused, we only selected a subset of the proposed metrics (e.g., we didn't analyze saccade precision or the psychometric Point of subjective equality).

To search for structure in the data, we performed the PCA on the isolated metrics alone since the interaction measurements were already directly related to multiple different saccadic and pursuit measurements. Instead of an additional multidimensional scaling approach, we explored the data with selected correlations to directly test the interaction of different sensory information and different oculomotor behaviors and to estimate the relative contributions of sensory and motor information. For studying saccade–pursuit interactions, we decided against the use of additional *t* tests to compare good and bad performers for each eye movement but stuck with the proposed correlational approach.

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